Qatar-1b

R-1844

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_150403` · created 2026-08-03T07:17:41+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457115.93816 +/- 0.00051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.02
Transit depth (Rp/Rs)^2	3.7 +/- 0.77 [%]
Orbital Inclination (inc)	83.76 +/- 1.85
Airmass coefficient 1 (a1)	0.987 +/- 0.03
Airmass coefficient 2 (a2)	0.002 +/- 0.024
Scatter in the residuals of the lightcurve fit is	3.08 %
Best Comparison Star	#3 - [400, 285]
Optimal Method	PSF photometry
Transit Duration (day)	0.069 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.02 vs 0.1463 (deviation 2.28σ)
Duration fit vs published	0.069 d vs 0.0692 d (deviation 0.01σ)
Mid-time O–C	-0.2 min
Depth SNR	9.6
Residual scatter	3.08 %

Light curve

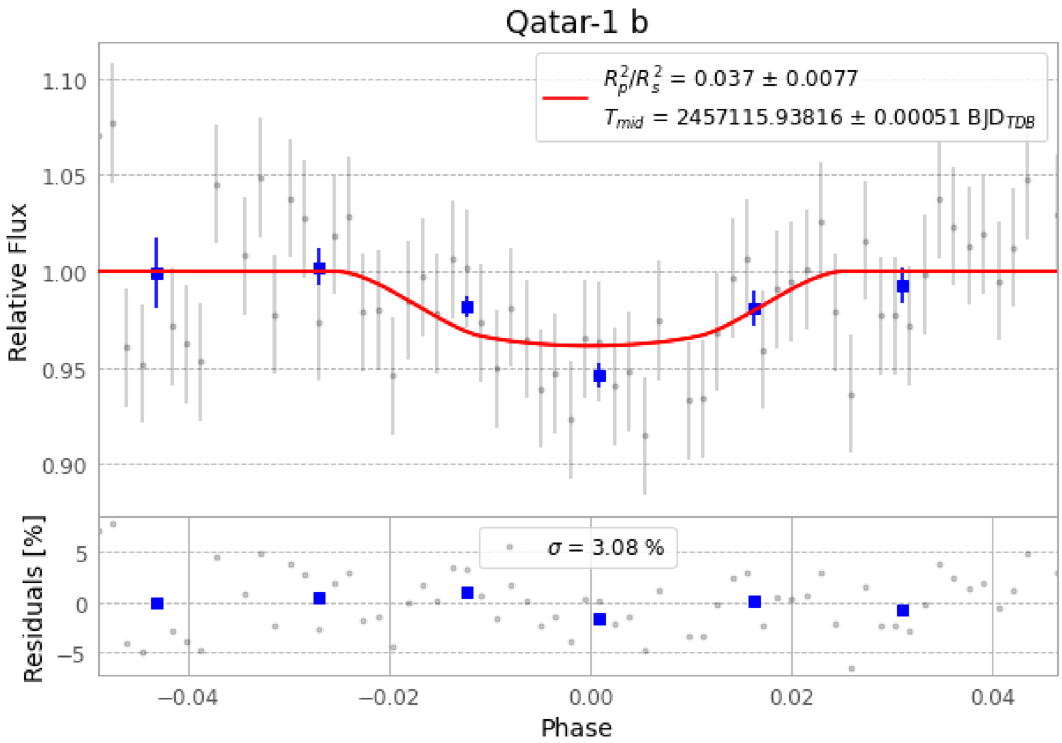

Figure 1 — FinalLightCurve_Qatar-1 b_2015-04-03.png (SHA-256 c17975776b057889…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 206], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 53a551e7131bf958…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

65 FITS frames (43 MB), `Qatar-1150403085112.FITS` → `Qatar-1150403120612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-150403.log (49ed2e04482c…), FinalLightCurve_Qatar-1 b_2015-04-03.png (c17975776b05…), FinalLightCurve_Qatar-1 b_2015-04-03.csv (48cb46ae1f3a…)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-03 07:17Z.